MATH1231 CALCULUS
Chapter 4C
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(4.4) Introduction

A **series** is the **sum of a sequence**.

E.g., summing the sequence $\{1, 2, 3, \ldots, 99, 100\}$

gives the series $\displaystyle\sum_{k=1}^{100} k = \sum_{i=1}^{100} i = 1 + 2 + 3 + \cdots + 99 + 100$.

Given an **infinite sequence** $\{a_n\}$, we define the **infinite series** $\displaystyle\sum_{k=0}^{\infty} a_k$.

Example: $\displaystyle\sum_{k=1}^{\infty} \frac{1}{2^k} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 1$ Proof: geometry!

In spite of summing an infinite number of terms, we get a finite result:
The series **converges**.

Convergence of Series

Does a given infinite series converge or diverge?
Construct the sequence $\{s_n\}$ of **partial sums**

$$
\begin{aligned}
s_1 &= a_1 \\
s_2 &= a_1 + a_2 \\
&\vdots \\
s_n &= a_1 + a_2 + \cdots + a_n = \sum_{k=1}^{n} a_k \\
&\vdots
\end{aligned}
$$

If the sequence of partial sums $\{s_n\}$ **converges** then the series **converges**, i.e.,

If $\lim_{n\to\infty} s_n = L$ then $\sum_{k=0}^{\infty} a_k = L$

If $\{s_n\}_{n=0}^{\infty}$ **diverges** then $\sum_{k=0}^{\infty} a_k$ **diverges**

Geometric series

$$\sum_{k=0}^{\infty} r^k, \qquad \text{where } r \in \mathbb{R}$$

$$\begin{aligned} S_n &= 1 + r + r^2 + r^3 \dots r^{n-1} + r^n \\ r\,S_n &= r + r^2 + r^3 \quad \dots \quad + r^n + r^{n+1} \end{aligned}$$

subtract and solve for S_n:

$$S_n - r\,S_n = 1 - r^{n+1} \qquad \Longrightarrow \qquad S_n = \frac{1 - r^{n+1}}{1 - r}, \qquad r \neq 1$$

Thus, we see that

(i) if $|r| < 1$ the series converges, and $\displaystyle\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$

(ii) if $|r| \geq 1$ the series diverges.

Proof

$$s_n = \sum_{k=0}^{n} r^k = \frac{1 - r^{n+1}}{1 - r} \quad \text{if} \quad r \neq 1$$

If $r = 1$ then $s_n = n + 1$ **diverges** as $n \to \infty$

If $r = -1$ then $\{s_n\} = \{1, 0, 1, 0, \ldots\}$ is **boundedly divergent**.

If $|r| < 1$ then $r^{n+1} \to 0$ as $n \to \infty$, so s_n **converges to** $\frac{1}{1-r}$

If $|r| > 1$ then r^{n+1} **diverges** as $n \to \infty$, so does s_n.

Telescoping series

Another type of infinite series which we can deal with is the so-called telescoping series. Consider, e.g., the series $\sum_{k=1}^{\infty} \frac{1}{k^2 + k}$

(4.5) Tests for convergence of series

There are many tests for the convergence (or divergence) of a given series. We will discuss the following in MATH1231:

- *nth*-term test for divergence
- integral test
- comparison test
- ratio test
- alternating series test
- absolute convergence test

The k^{th}-term Test for Divergence

Theorem

If $\displaystyle\sum_{k=0}^{\infty} a_k$ converges then $a_k \to 0$ as $k \to \infty$.

Proof

Suppose $\displaystyle\sum_{k=0}^{n} a_k \to L$ and define $s_n = \displaystyle\sum_{k=0}^{n} a_k$

Then $s_n - s_{n-1} = a_n \Rightarrow \displaystyle\lim_{n\to\infty} a_n = \lim_{n\to\infty} s_n - \lim_{n\to\infty} s_{n-1} = L - L = 0$

kth Term Test

$$\boxed{\text{If } a_k \not\to 0 \text{ as } k \to \infty \text{ then } \sum_{k=0}^{\infty} a_k \text{ diverges.}}$$

Be careful! You cannot use this test to show that a series converges. The test is only useful for showing that a series *diverges*.

Examples

(i) Does $\sum_{n=1}^{\infty} \frac{n}{2n+3}$ converge?

(ii) Does $\sum_{k=0}^{\infty} (-1)^k$ converge?

What is wrong with the following? Let

$$S = \sum_{k=0}^{\infty} (-1)^k = 1 - 1 + 1 - 1 + \dots$$

Thus,

$$\begin{aligned} -S &= -1 + (1 - 1 + 1 - 1 + \dots) && (1) \\ &= -1 + S && (2) \end{aligned}$$

Solving for S suggests $S = 1/2$ . . . .

Linearity of Convergent Series

$a_k \in \mathbb{R},\ k \in \mathbb{N}$

If $\sum^{\infty} a_k$ and $\sum^{\infty} b_k$ both converge then

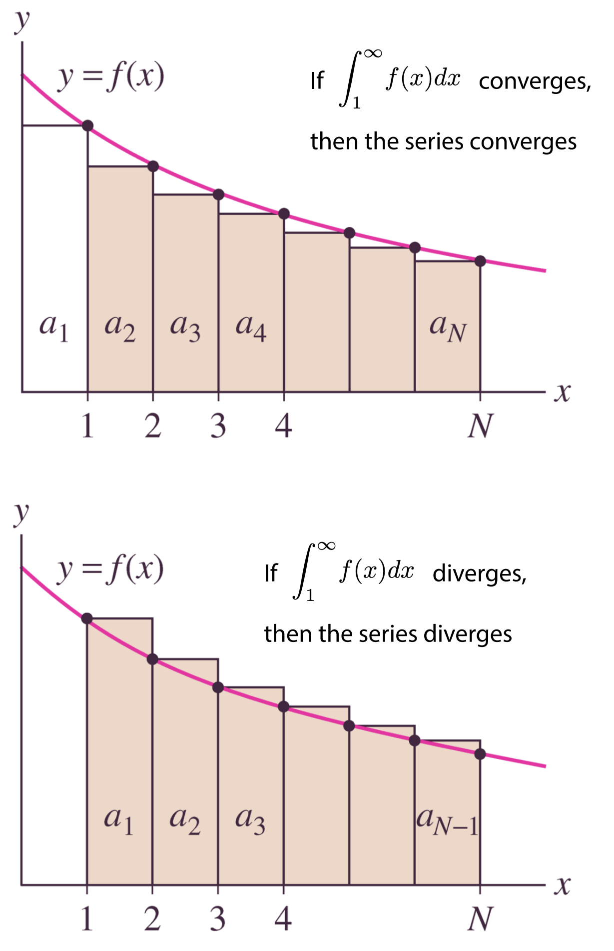
$$\sum^{\infty}(a_k + b_k) = \sum^{\infty} a_k + \sum^{\infty} b_k \quad \text{and} \quad \sum^{\infty}(\alpha a_k) = \alpha \sum^{\infty} a_k$$

Note 1: $\sum^{\infty}(a_k + b_k)$ can converge without either of $\sum^{\infty} a_k$ or $\sum^{\infty} b_k$ converging: consider $\quad a_k = k, \quad b_k = -k.$

Note 2: $\sum_{k=0}^{\infty} a_k$ converges if and only if $\sum_{k=N}^{\infty} a_k$ converges

The Integral Test

Theorem: Let $a_n > 0, \ \forall n$ and $a_n = f(n)$, where $f(x)$ is **continuous, positive, decreasing** $\forall x \geq N$. Then $\sum_{n=N}^{\infty} a_n$ and $\int_N^{\infty} f(x)\,dx$ both converge or both diverge.



Example

Show that the Harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

Example

Show that the series $\sum_{n=1}^{\infty} \frac{n}{(n^2+1)^2}$ converges.

Example

Show that $\displaystyle\sum_{k=2}^{\infty} \frac{1}{k \ln k}$ diverges.

The p-Series

$$\sum_{k=1}^{\infty} \frac{1}{k^p} \qquad \begin{array}{ll} \text{converges if} & p > 1 \\ \text{diverges if} & p \leq 1 \end{array}$$

Proof

i) For $p = 1$ this is the divergent Harmonic series (integral test)

ii) For $p \neq 1$ use the integral test again:

Examples

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^{-1}}$$

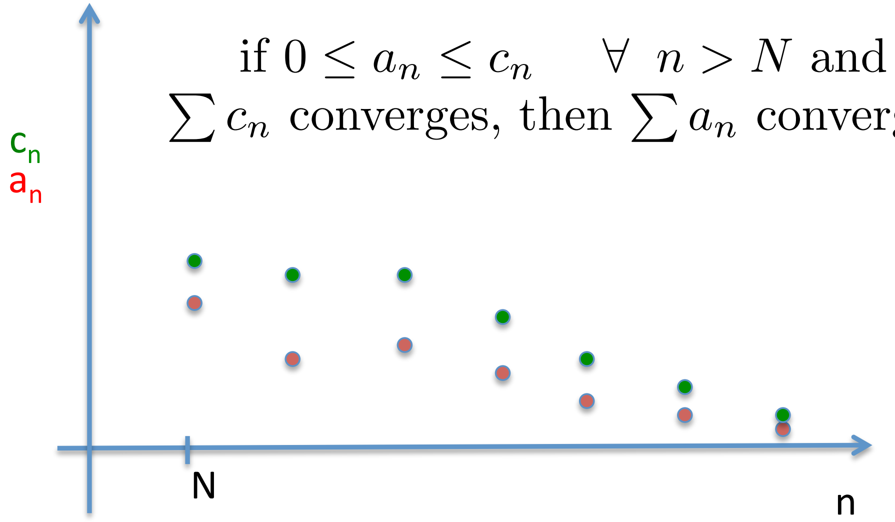
$$\sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^{7/6}}$$

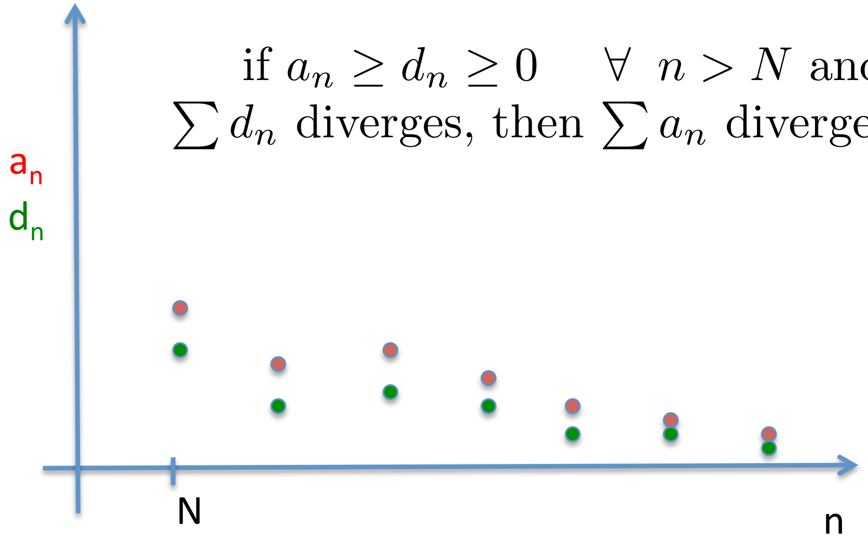
Comparison Test

Comparison test for series convergence:

if $0 \leq a_n \leq c_n \quad \forall \; n > N$ and
$\sum c_n$ converges, then $\sum a_n$ converges.



if $a_n \geq d_n \geq 0 \quad \forall \; n > N$ and
$\sum d_n$ diverges, then $\sum a_n$ diverges.



Example

Does the following series converge? $\displaystyle\sum_{k=1}^{\infty} \frac{k}{k^3+1}$

Example

Does the following series converge? $\displaystyle\sum_{k=2}^{\infty} \frac{1}{\sqrt{k^2-1}}$

d'Alembert's ratio test

Suppose $a_k > 0$ and $\frac{a_{k+1}}{a_k} \to r$ as $k \to \infty$.

Then $\sum^{\infty} a_k$ **converges** if $r < 1$ and **diverges** if $r > 1$.

- If $r = 1$ the ratio test is inconclusive (need another approach)
- The ratio test works well for k factorials.

Example: Does the series $\displaystyle\sum_{k=117}^{\infty} k^2 \frac{2^k}{3^k}$ converge?

Example

Does the series $\sum_{n=0}^{\infty} \frac{n^2}{n!}$ converge?

Example

Does the series $\sum_{n=0}^{\infty} \frac{(2n)!}{n!n!}$ converge?

Example

Does the series $\displaystyle\sum_{n=0}^{\infty} \frac{4^n n! n!}{(2n)!}$ converge?

Alternating Series

Definition

An **alternating series** is one in which the terms are alternately positive and negative.

Example

$\sum \frac{(-1)^k}{k}$ is an **alternating series**

4.5.4 Leibniz Alternating Series Test

If

(i) $a_0 \geq a_1 \geq a_2 \geq \cdots \geq a_k \geq \cdots \geq 0$ and

(ii) $a_k \to 0$ as $k \to \infty$,

then

the alternating series $\displaystyle\sum_{k=0}^{\infty} (-1)^k \, a_k$ **converges**.

Example

Show that $\displaystyle\sum_{k=1}^{\infty} \frac{(-1)^k}{k}$ converges.

Example

Does the series $\displaystyle\sum_{k=2}^{\infty}(-1)^{(k-1)}\frac{2k}{4k^2-3}$ converge?

Absolute Convergence

Definition:

$$\sum_k^{\infty} a_k \textbf{ converges absolutely} \text{ if } \sum_k^{\infty} |a_k| \textbf{ converges}.$$

(For a series of positive terms, this is the same as ordinary convergence.)

Theorem:

If $\displaystyle\sum_k^{\infty} |a_k|$ converges, then $\displaystyle\sum_k^{\infty} a_k$ converges.

Example

Show that $\displaystyle\sum_{n=1}^{\infty} \frac{\sin n}{n^2}$ converges absolutely.

Does $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ converge absolutely?

Conditional Convergence

Definition:

A series is **conditionally convergent** if it is convergent but not absolutely convergent.

Example:

$\displaystyle\sum_{k=1}^{\infty} \frac{(-1)^k}{k}$ is **convergent** but $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k}$ is **divergent**

Example

Show that the series $\displaystyle\sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k}}$ is conditionally convergent.

Rearrangements of Series

When can we add up all the terms of a series in a different order but still get the same result?

Theorem:

(i) If $\sum_{k}^{\infty} a_k$ is **absolutely convergent** then no matter in which order you add up the terms, you get the same limit.

(ii) If $\sum_{k}^{\infty} a_k$ is **conditionally convergent**, then, given any $L \in \mathbb{R}$, we can **find** a rearrangement of the terms a_k so that the new series has limit L. Moreover, every conditionally convergent series has a re-arrangement that diverges to ∞, and another rearrangement that diverges to $-\infty$.

Notes

1. $\sum a_k$ is **absolutely convergent** if and only if every rearrangement of $\sum a_n$ converges to the same limit.
2. Never rearrange the terms of a **conditionally convergent** series to find the limit.